Sales Data Analysis — Short Report

CodeOrbit Tech, Data Analyst Internship — Task 2: Sales Data Analysis with Pandas Dataset: Cleaned sales data from Task 1 (`data/Cleaned_Data.csv`, 466 orders, Jan–Sep 2026) **Tools:** Python, pandas, matplotlib

Key metrics

- Total orders: **466**
- Total recorded sales (all statuses): **316,910.42**
- Average order value: **680.07**
- Total quantity sold: **824 units** (average 1.77 per order)
- Date range: **2 Jan 2026 to 27 Sep 2026** (25 orders have no date and are excluded only from the monthly analysis)

Top products

Product	Total sales	Orders	Average order value
Laptop	144,470.78	81	1,783.59
Phone	74,203.84	70	1,060.05
Monitor	41,374.69	79	523.73
Tablet	33,209.15	46	721.94
Headphones	10,054.00	56	179.54
Keyboard	6,411.65	66	97.15
Unknown	3,997.72	6	666.29
Mouse	3,188.59	62	51.43

Laptop alone is 45.6% of recorded sales. Mouse and Keyboard are bought often but cheap, so they add little sales value.

Sales by city

City	Total sales	Orders
Rawalpindi	71,229.34	88
Faisalabad	61,216.48	91
Karachi	58,742.36	86
Islamabad	52,692.50	92
Lahore	49,412.46	87
Unknown	23,617.28	22

Sales are spread across cities — no single city dominates.

Sales by month

June 2026 was the strongest month (55,415.64, 62 orders) and April 2026 the weakest (21,506.54, 42 orders). Sales rise and fall month to month rather than growing steadily.

Payment and status — read the headline total carefully

- Top payment method by sales: Credit Card (81,167.63), then Easypaisa (53,311.61) and JazzCash (52,927.51).
- By status: Cancelled 89,730.34 (126 orders, 28.3% of recorded sales), Completed 86,842.34 (124 orders), Pending 52,424.89, Shipped 51,180.82, Returned 22,673.83.
- **Completed-only sales are 86,842.34.** The headline total of 316,910.42 includes cancelled, pending and returned orders, so it is not the same as money actually earned.

Findings

1. High-value products (Laptop, Phone) drive sales; low-price products drive order counts.
2. Demand is geographically spread, with Rawalpindi slightly ahead.
3. Monthly sales are uneven — June peaked, April dipped.
4. The high cancellation rate (28.3% of recorded sales) is the biggest issue to investigate next.

Limitations

- Small, synthetic sample dataset — findings describe this sample only.
- 25 orders lack a date, so monthly totals cover 441 dated orders.
- Some values remain "Unknown" where the original data could not be recovered in Task 1.